WANG XIN

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Education

2018.09 — Pres. **M.Eng.**, Faculty of Artificial Intelligence in Education, Central China Norma University (CCNU)
Supervisor: Qiusha Min

2016.02 — 2016.07 **Exchange Student**, College of Science, Providence University (PU)

2014.09 — 2018.07 **B.Eng.**, Zhang Zhongjing College of Chinese Medicine, Nanyang Institute of Technology (NYIST)

Research Interests

- My researches focus on machine learning, deep learning, and medical imaging;
- My research interest lies at the intersection of machine learning, deep learning, computer vision, medical image compression, and medical image segmentation.

Research Experience

Faculty of Artificial Intelligence in Education, Central China Normal University
Master student **2018.09 — Pres.**

- **Deep Learning and Medical Image Compressions**: The recent research explore new methods for lossless compression of medical data, leveraging image segmentation and deep learning [2,3]. This technique first divides the medical image data into specific regions based on anatomical features, namely, image density, relative position, and organ size. A deep neural network is then trained to generate a series of optimal predictors for each region. These predictors can be adaptively switched based on the characteristics of the region being compressed. Instead of relying on a global prediction model, the proposed technique would adaptively switch to an optimal predictor according to the characteristics of the region being compressed. (Fall 2020).
- **Medical Image Processing**: Different technologies (Oracle Java, Adobe Flash, and HTML5) are used to develop a web-based medical imaging application [1]. Simultaneously, this application connects to a medical image server and can provides several required functions for radiological interpretation, e.g. navigation, magnification, windowing, maximum intensity projection, fly-through, surface rendering, and volume rendering. Java-, Flash-, and HTML5-based medical imaging applications were tested on different operating systems over a local area network and a wide area network. (Fall 2019).
- **Machine Learning and Deep Learning**: In addition to the machine learning courses in the teaching plan, I also learned machine learning and deep learning on the Coursera online platform (Fall 2018).
  - Machine Learning on the Coursera platform provided by Stanford University
    Lectures: Andrew Ng
  - Deep Learning on the Coursera platform provided by Stanford University
    Lectures: Andrew Ng, Teaching Assistant-Kian Katanforoosh & Teaching Assistant-Younes Bensouda Mourri
    (1) Neural networks and deep learning
    (2) Improving deep neural networks: hyperparameter tuning, regularization and optimization
    (3) Structuring machine learning projects
    (4) Convolutional neural networks
    (5) Sequence models

Teaching Experiences

Course Coordinator, September 2018 – January 2019.
Central China Normal University Wollongong Joint Institute, Reasoning and Learning.

- Led the lab session for 90 students in Multimedia Technologies class. (Fall 2018)

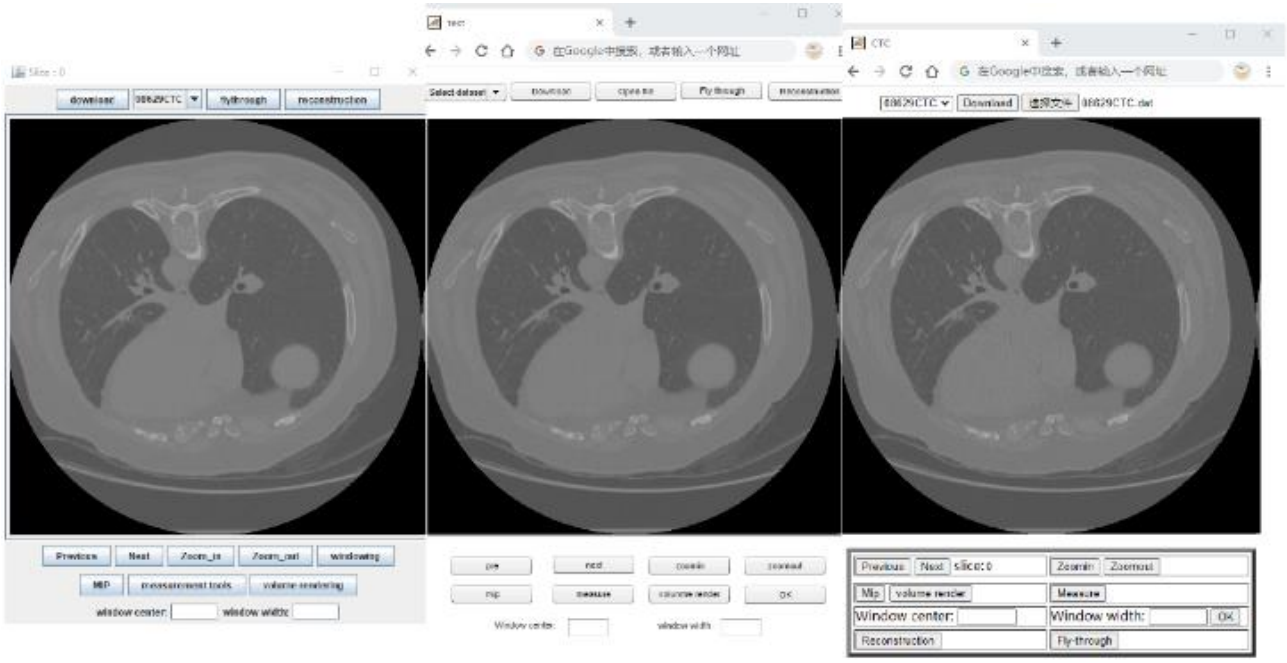
Teaching Assistant, September 2019 – January 2021.
Central China Normal University Wollongong Joint Institute, Reasoning and Learning.

- Teaching assistant for the graduate-level course CSCI924_JI119 – Reasoning and Learning, which is oriented to master students in the Central China Normal University Wollongong Joint Institute, Central China Normal University. (Fall 2019 & Fall 2020)

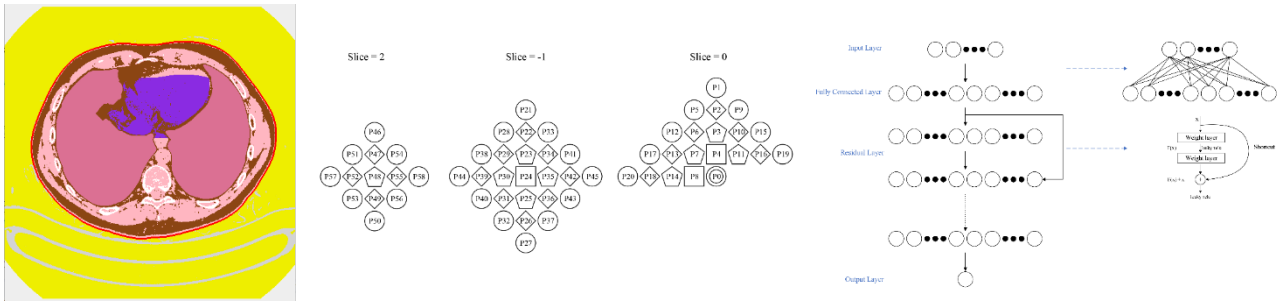
Publications

[1] Min Q*, Wang X*, Huang B, et al. Web-Based Technology for Remote Viewing of Radiological Images: App Validation[J]. Journal of Medical Internet Research, 2020, 22(9): e16224. (SCI: 5.03)

*these authors contributed equally



[2] Qiusha Min, Xin Wang, Bo Huang. Lossless medical image compression based on region segmentation and residual neural network, 2021. (Under Review)



[3] Qiusha Min*, Neng Liu*, Xin Wang*. Lossless medical image compression based on region segmentation and decision tree, 2021. (In Preparation)

Patents & Software Copyright

[4] Remote medical image access system based on Java technology, Qiusha Min, Xin Wang, Software copyright, 2019SR0982502
[5] Remote medical image access system based on HTML5 technology, Qiusha Min, Xin Wang, Software copyright, 2019SR0982496
[6] Remote medical image access system based on Flash technology, Qiusha Min, Xin Wang, Software copyright, 2019SR0982505

Skills

- Computing Skills: Python, MATLAB, Java, TensorFlow, Keras, PyTorch.
- Language: Chinese (native), English (fluent).
- Machine Learning (Coursera), Deep Learning (Coursera), AI for Medicine (Coursera), Object oriented programming-Java (MOOC).